

Biology
BIOLOGY 3ET3
Ecotoxicology
Winter 2027



SCIENCE

Instructor Information

Joanna Wilson

Email: wilsojo@mcmaster.ca

Office Hours:

TBD, on teams

Course Information

Lectures:

Wednesday 9:30 am- 10:20 pm

Fridays 9:30 am- 10:20 am

Please check your mosaic calendar for location

Labs: NA

Tutorials: Tuesdays 4:30 pm – 6:20 pm

Course Dates: 01/04/2027 - 04/07/2027

Units: 3.00

Course Delivery Mode: In Person

Course Description: This course covers the environmental pathways of exposures of terrestrial and aquatic organisms to modern and legacy contaminants, and the adverse effects that these contaminants have on diverse species including humans. Lectures (two hours), one tutorial (two hours); one term Prerequisite(s): CHEM 1A03, 1AA3 and registration in Level III or above. Completion of BIOLOGY 2A03, 2B03, 2F03 and one of CHEMBIO 20A3, 20C3 (or 20G3), CHEM 2E03, 20A3 (or 20G3), ENVIRSCI 3003 are recommended.

Instructor-Specific Course Information

Communication between Students & Instructional Team: Dr. Wilson and the TAs will respond to queries about lectures and tutorials on Teams. Lecture questions should be posted to the Lecture channel instead of through email so that everyone can see the question and answer. Tutorial questions can be posted to the channel for your specific tutorial section.

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)
- [Open Education Resources](#)

Course Learning Outcomes

- • Appreciate the mechanisms of toxicity in organisms, populations and communities
- • Understand the classes of contaminants, their fate in the environment and routes of organism exposure
- • Use the breadth of topic knowledge from lectures and readings to answer fact-based and applied questions
- • Conduct basic exposure, population and community ecotoxicological assessments
- • Read, understand and discuss journal articles relating to ecotoxicology studies

Course Learning Goals

- This course covers the sources and fate of contaminants in the environment, and the adverse effects that these contaminants have on diverse species including humans. Topics will include: major classes of contaminants; contaminant sources and environmental fate; exposure routes in organisms; biotransformation; bioavailability, bioaccumulation, and biomagnification;

toxicity tests; molecular, cellular, tissue and individual effects of contaminants; population & community effects of contaminants; global patterns of contaminants in humans and wildlife; biomonitoring and field-based testing; ecological risk assessments; whole ecosystem studies; toxicity testing; current issues in ecotoxicology. This course will be organized into weekly lectures and tutorials.

Class Format

In Person

In Person

IMPORTANT NOTE: Each week is a mix of lectures and tutorials for a maximum of 4 hours per week. Some weeks there will only be 2 hours of lectures, other weeks there will be 2 hours of lectures and 2 hours of tutorial. Additional material is posted on Avenue to Learn for asynchronous learning. A detailed schedule has been posted on Avenue to Learn and below.

The lectures will be a synthesis of several sources of information: a textbook (not required for students, available at the library) Fundamentals of Ecotoxicology and scientific literature. Beyond the lectures, materials (e.g. videos, other materials) will be posted on Avenue to Learn, under course content. Students are responsible for learning all material (lectures and posted materials for asynchronous learning) related to the course.

Tutorial materials will be posted the week before each tutorial. Tutorial sessions will enable students to learn practical skills in ecotoxicology and explore lecture materials in more depth through discussions and assignments. Tutorials will take place in person most weeks throughout the semester, as per the schedule posted on Avenue to Learn, and will include an introduction at the beginning (recorded), in-tutorial exercises and post-tutorial assignments. All tutorial assignments will require a computer to complete. Tutorial time will be used to work on assignments and ask questions, and this part will not be recorded.

Tutorials with an assignment will be part of the midterm or final exam materials. All tutorial assignments will be submitted to electronic drop boxes on Avenue by 5 pm on the due date.

While we expect the course to be delivered in person, virtual lectures may be provided as needed and classes will be recorded so that students who miss classes due to illness have

access to lecture material. Pedagogically, the recordings will allow students to re-visit and review presented materials more than once as a study tool and will enable inclusive access for SAS-registered students who require closed captioning of course lectures.

These are not meant to replace in class learning and students will do better in the course if they use the recorded lectures to supplement, and not to replace, in person learning.

Please note that students are expected to attend all lectures and tutorials and study all materials posted on the Biology 3ET3 Avenue to learn course website. Students will benefit most by including their own detailed written or typed notes in addition to the provided lecture and tutorial resources and slides posted on Avenue to Learn.

Course Evaluation

Final marks for the course are based on a total assessment of each student's record. It is a student's responsibility to make sure that their marks are complete and correct. If there are any discrepancies, students must contact the Biology 3ET3 Course Instructor via email at joanna.wilson@mcmaster.ca before the final exam.

Marks will NOT be bell-curved at any point in the term. The Biology Department does not approve of altering marks arbitrarily at a student's request. Refer to the Biology 3ET3 Course Policies below for

information regarding regrading of assignments. Students are expected to abide by the University's academic integrity expectations when completing all assessments in Biology 3ET3 in accordance with the following statement: By submitting this work, I certify that the work represents solely my own independent efforts. I confirm that I am expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. I confirm that it is my responsibility to understand what constitutes academic dishonesty under the Academic Integrity Policy.

Assessment	Weight	Details
Tutorial Assignments	45%	Assigned in tutorials, 3 assignments, each worth 15%, due ~2-3 weeks after the tutorial associated with that assignment
Mid Term	25%	Feb (near 27) during tutorial time slot (location TBD) A make-up test will be provided on (give date) during tutorial time slot (location TBD)

Exam	30%	The Final Exam will take place during the April Final Exam period, as Scheduled by the registrar's office. It will cover the entire course but with stronger emphasis on materials not covered in the Midterm
Total Marks	100%	

Course Evaluation Details

There are three assignments throughout the semester, each worth 15%, and due 2-3 weeks after the associated tutorial. Details on their due dates are in the course schedule herein and are also posted as a pdf on Avenue to Learn.

The midterm and final exam will be administered in person at locations on the McMaster campus. The midterm and final exam will be completed individually and are NOT open book. All questions may evaluate factual, conceptual, and application knowledge. Both the midterm and final exam will consist of multiple-choice questions as well as short answer questions. Details on the content and format will be provided to you well in advance of the dates of the exams.

Midterm : The midterm is 90 minutes in duration and will be held during regularly scheduled class time on Tuesday Feb 23rd. It will cover materials up to and including February 12th (week 6), including lecture-related resources posted on the course website and tutorial materials (T1-T4).

Final Exam: The final exam will be 2.5 hours in duration. The final exam will evaluate course information covered during the entire academic term including lecture-related resources and tutorial materials that are not part of the three assignments. There will be an emphasis on material covered after February 13th (week 7 onward). It will be held during the Final Exam period and will be scheduled by the Registrar in February.

It is the responsibility of the student to attend and study all lectures and attend all tutorials. If a tutorial or a lecture is missed, students are responsible for the

covered material.

It is an expectation that students complete at least 75% of the course work, including tutorials, exams, and assignments, to obtain credit for Biology 3ET3.

It is the responsibility of the student to attend all tutorials as scheduled. If a student is unable to complete the post-tutorial assignment, a grade of zero will result unless the student's absence is supported by a self-reported McMaster Student Absence Form (MSAF) or one from the student's Associate Dean's Office. Refer to the Biology 3ET3 MSAF policy regarding missed tutorial assessments.

The Midterm and Final Exams will be completed in person, at locations on the McMaster campus. Only use of a McMaster University approved calculator (Casio fx-991MS or Casio fx-991MS Plus) is allowed during evaluations. Written answers must be completed in blue or black ink pen. Use of correction fluid or tape is not permitted on tests or exams.

All exams and post-tutorial assignments should be completed and submitted individually unless other instructions to work in groups are specifically defined. All reports and assignments which are submitted should be unique. It is considered academic dishonesty to submit work that was not originally yours or that has been previously submitted. All cases of academic dishonesty will be dealt with through the Office of Academic Integrity at McMaster University.

Any marked term work (assignments, exams, etc.) may be submitted for re-grading within 5 business days of the work being returned to the student. The request for a regrade should be emailed to both Dr. Wilson and the teaching assistant and include details on the reason for the regrade. Regrade requests made for frivolous reasons will be denied.

Any term mark corrections on Avenue to Learn must be made BEFORE the Biology 3ET3 Final Exam in April 2025. Contact the Course Instructor via e-mail (joanna.wilson@mcmaster.ca) regarding midterm or assignment grade corrections. There are no alternative assignments that can be completed by students for the purpose of increasing their final grade.

For the purposes of the late withdrawal policy, the exam is considered the final assessment of this course.

Undergraduate Grading Scale

The grades breakdown for McMaster's 12-Point Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Graduate Grading Scale

Graduate Student Grading Scale (Except for MBA and Master of Finance)

Grade	Points	Equivalent Percentage	Pass/Fail
A+	12	90-100	P+
A	11	85-89	P
A-	10	80-84	
B+	9	77-79	
B	8	73-76	
B-	7	70-72	
F	0	69 and under	F

MBA and Master of Finance Grading Scale

Grade	Points	Equivalent Percentage	Pass/Fail
A+	12	90-100	P+

Grade	Points	Equivalent Percentage	Pass/Fail
A	11	85-89	P
A-	10	80-84	
B+	9	75-79	
B	8	70-74	
B-	7	60-69	
F	0	59 and under	F

Course Schedule

Week	Topics
1 – Jan 6,8	Introduction to ecotoxicology Major classes of contaminants 1. No Tutorials this week
2 – Jan 13,15	Sources and fate of chemicals Contaminant uptake, biotransformation, storage and elimination 1. Tutorial 1: Data analysis and graphing
3 – Jan 20,22	Bioconcentration, bioaccumulation and biomagnification in food chains Molecular effects 1. Tutorial 2: Exposure assessment – assignment 1
4 – Jan 27,29	Tissue effects Individual responses to contaminants 1. No tutorial
5 – Feb 3,5	Population responses to contaminants Community responses to contaminants • Tutorial 3: Human biomonitoring in Canada • Assignment 1 Due
6 – Feb 10,12	Toxicity testing Regulatory toxicity testing • Tutorial 4. Community effects – Assignment 2

7 – Feb 17,19	<u>READING BREAK</u>
8 – Feb 24,26	Field approaches used in Ecotoxicology No lecture • Midterm in tutorial slot, materials up to and including (week 6)
9 – March 3,5	Current issues: Omics in ecotoxicology Current issues: Pharmaceutical pollution • Tutorial 5: Toxicity databases
10 – March 10,12	Current issues: Mercury Current Issues: Microplastics • Tutorial 6: Microplastics
11 – March 17,19	Current issues – Environmental racism/justice Current issues – Chemical contamination from E-waste • Tutorial 7: Environmental racism case studies
12 – March 24.26	Current issues – New Approach Methods in Toxicity Testing Current issues – Adverse Outcome Pathways • Make up midterm, in tutorial slot, location TBD
13- March 31, Apr 1	Current Issues: Species choice in ecotoxicology testing Current Issues : TBD • Tutorial 8: Class presentations – Assignment 3- Gr
14 – Apr 7	Review for Exam • Tutorial 9: Class presentations – Assignment 3- Gr

Current issues could include endocrine disruptors, pharmaceuticals, road salts etc and may include some guest lectures.

Midterm will cover all the lectures and tutorials up to, and including, week 6.

There will be a **make-up midterm** offered for students who miss the midterm, if they have an approved MSAF.

The **Exam** will cover all the lectures and tutorials in the course, with an emphasis on materials after week 6 topics.

Late Assignments

If no MSAF is approved and the assignment is submitted late, a penalty of 10% of the total marks for that assignment per day will apply (e.g. 4 marks out of 40 for each day it is late).

Absences, Missed Work, Illness

[McMaster Student Absence Form \(MSAF\)](#): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar “Requests for Relief for Missed Academic Term Work”.

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster’s use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Course Modification

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If any modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes. Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.

Generative AI: Use Prohibited

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it “shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source”. This includes work created by generative AI tools. Also stated in the policy is the following, “Contract Cheating is the act of “outsourcing of student work to third parties” (Lancaster & Clarke, 2016, p. 639) with or without payment.” Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

Course Specific Relief for Missed Academic Work Information

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Biology 3ET3 MSAF Policy

Course component not completed	Result
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Midterm (25% of final grade)	There is a make-up midterm provided. With an approved MSAF, you may write the make-up midterm. If no MSAF is approved, a grade of zero will be recorded for the missed midterm.
Tutorial Assignments (15% each of final grade)	With an approved MSAF, you will automatically have a 72-hour extension on the original deadline. It is important to email Dr. Wilson and your TA about your MSAF accommodations. If no MSAF is approved and no assignment is submitted, a grade of zero will be recorded for the missed assignment. If no MSAF is approved and the assignment is submitted late, a penalty of 10% of the total marks for that assignment per day will apply (e.g. 4 marks out of 40 for each day it is late).

Checking Grades & Regrade Policy

Checking Your Grades

Your marks will be recorded on Avenue. It is your responsibility to check that all grades entered into Avenue are recorded properly. All grade concerns and discrepancies must be reported to the Instructional Assistant within a week of receiving the grade.

Regrade Policy

Office hours with the TAs (and or instructor/IA) will be arranged following assessments to address concerns students may have regarding their grade. Requests for re-evaluation of tests or assignments must be made in writing to the IA using the form provided in Avenue and justifying the reason for which the assignment should be remarked. This should be done within one week of discussing the marked test or assignment with the instructional team and BEFORE the final exam is written. If the IA considers the written justification to be

insufficient (e.g. simply wanting a higher grade is insufficient), the assignment will not be re-graded. If the justification is considered sufficient, the entire assignment/test will be re-graded. Please be aware that this may result in an increase, decrease or no change to the original mark. There will be no alternative assignments that can be completed by students for the purpose of increasing their final grade.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#).

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to

Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, Indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.